

hw 0427

12.4

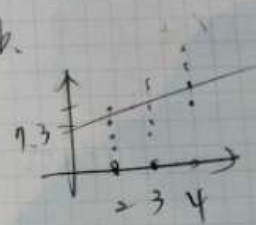
- ③ 1st 圖: - 直線 $\Rightarrow$ 正態分配
 2nd 圖: 扇形 $\Rightarrow$ Fixed value $\propto$ Var
 可能 violate constant variance assumption
 $\Rightarrow$ Normality 尚可,
 equal variance 違反

- ④ a. $E(\epsilon) = 0$, 常態分佈.
 α 固定, ind
 b. $\sigma^2 = MSE = 58.05$
 c. Normal plot 接近直線
 Residual plot 無明顯型態
 有少數離群點
 $\Rightarrow$ Regression model assumption 成立

⑦

- a. $r^2 = 0.000729$.
 指出 almost no linear relationship
 b. normality or equal variance
 無明顯違反
 c. 線性度假設違反

12.5

- ② a. $\hat{y} = 3.6 + 12x$ d. $H_0: \beta = 0$
 $H_a: \beta \neq 0$
 $t = 4.39$
 $p < 0.01$, reject H_0
 b. 
 e. $0.001 < p < 0.01$
 f. 無違反回歸假設
 g. $\bar{x} = 3$ 95% C.I.
 $\hat{y} = 1.2$ (6.3, 8.09)
 c. $S^2 = MSE = 1.4$

⑮ a. $X = -1, \hat{y} = 2.95$
 99% C.I. = (0.93, 4.97)

b. $X = 1, \hat{y} = 0.95$
 99% P.I. = (-2.4, 4.3)

c. $X = \bar{x} = 0$

⑧ a. 有正相關線性關係

b. $\hat{y} = 70.8x + 80.9$

c. $p\text{-value} = 0.003$
 $0.003 < 0.05$, reject H_0

d. $X = 0.2$
 90% C.I. = (112.13, 157.9)

12.6

⑬ $H_0: \rho = 0$
 $H_a: \rho > 0$

$r = 1.329$
 $t = 0.92$

90% $\Rightarrow t_{0.05, 9} = 1.895 > 0.92$

Fail to reject $H_0 \Rightarrow$ 沒有充足 evidence to show 正相關

⑭

a. 用 correlation analysis

b. $r = 1.919$
 $\Rightarrow$ 強正相關

⑮

a. $r = 0.174$

b. $H_0: \rho = 0$ $t = 0.58$

$H_a: \rho = 0$ $t_{0.05, 10} = 2.228 > 0.58$

Fail to reject H_0